
EchoBreaker

How People Form and Update Beliefs About
Public Opinion During Social Media Browsing

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Motivation

- What you see on social media platforms (i.e. Twitter/X) is usually not representative of general public opinion.
 - More emotionally charged posts are often favored in algorithms.
 - People who post on social media tend to also have more polarizing opinions.
- People generally understand that social media is not representative, but still are (sometimes unconsciously) influenced by it. People start to believe the world is more divided than it actually is which has negative impacts.
- I am interested in tools that could ***reduce perception gaps***.

EchoBreaker

- A Chrome extension that lets you select a post on Twitter or Reddit and see what public opinion is on that post.
- The public output opinion is done through LLM synthetic surveys. Essentially, we create many different personas based on US census data (i.e. “30 year old, Asian, male, lives in San Francisco, income of 200k, etc.). Then the LLM asks each person about the social media post.
- This is supposed to mimic polling data. But obviously is much cheaper and almost instant.

Not really a ML or SWE Research Problem

- Most of student's research was a ML and/or SWE problem.
- EchoBreaker is an HCI (Human-centered interaction) problem. Specifically, it belongs in the “social computing” subfield of HCI.
- Therefore, I am not necessarily doing an evaluation against an existing benchmark to see if my tool is better than existing tools; rather, I will interview participants and see how respond to EchoBreaker.

Research Questions:

RQ1: What influences a person's perception of public opinion ***at the exact moment*** they see a social media post? Do the engagement metrics (like count, reply count) or contextual information (the tone of replies) matter?

RQ2: How do people respond when EchoBreaker presents data that contradicts their perception? Are they accepting or combative?

RQ: Any design guidelines for other researchers to build off this work?

DEMO

<https://www.loom.com/share/d5e5b020c8cb4cda8385083a3f869482>

Study Design

- I interviewed 3 participants for 40 minutes to an hour each.
- I try to get people who have varying political affiliations, age, gender, geolocation, etc.
- 2 were from my own network, one was recruited from Prolific (a platform to help researchers find participants).
- I asked them to use EchoBreaker on 3 pre-selected tweets.

Main Findings

- The gamification of LLM Synthetic Surveys (“Research version”) made it more enjoyable for participants to use. Also, it caused them to be more reflective which influenced them to be more open about updating their beliefs. This was perhaps my biggest finding and it was surprising!
- I expected a lot of pushback on LLM synthetic surveys. “I know better than AI”, “synthetic surveys can’t be trusted.” Instead, I was surprised that people were receptive to this concept. The more you rely on LLMs in general, the more you were accepting of EchoBreaker’s results. So at least in my N=3 study, everyone was very accepting in updating their beliefs and admitting if their public perception were wrong. That was also surprising!
- Contextual information on a social media post (like count, reply count, tone of replies, etc.) did not matter much in forming public opinion. The only thing that mattered was the person’s pre-existing opinions on the topic.
- Caveat: Small sample size. I plan to continue with this study after class and will interview more participants.

Thank You!